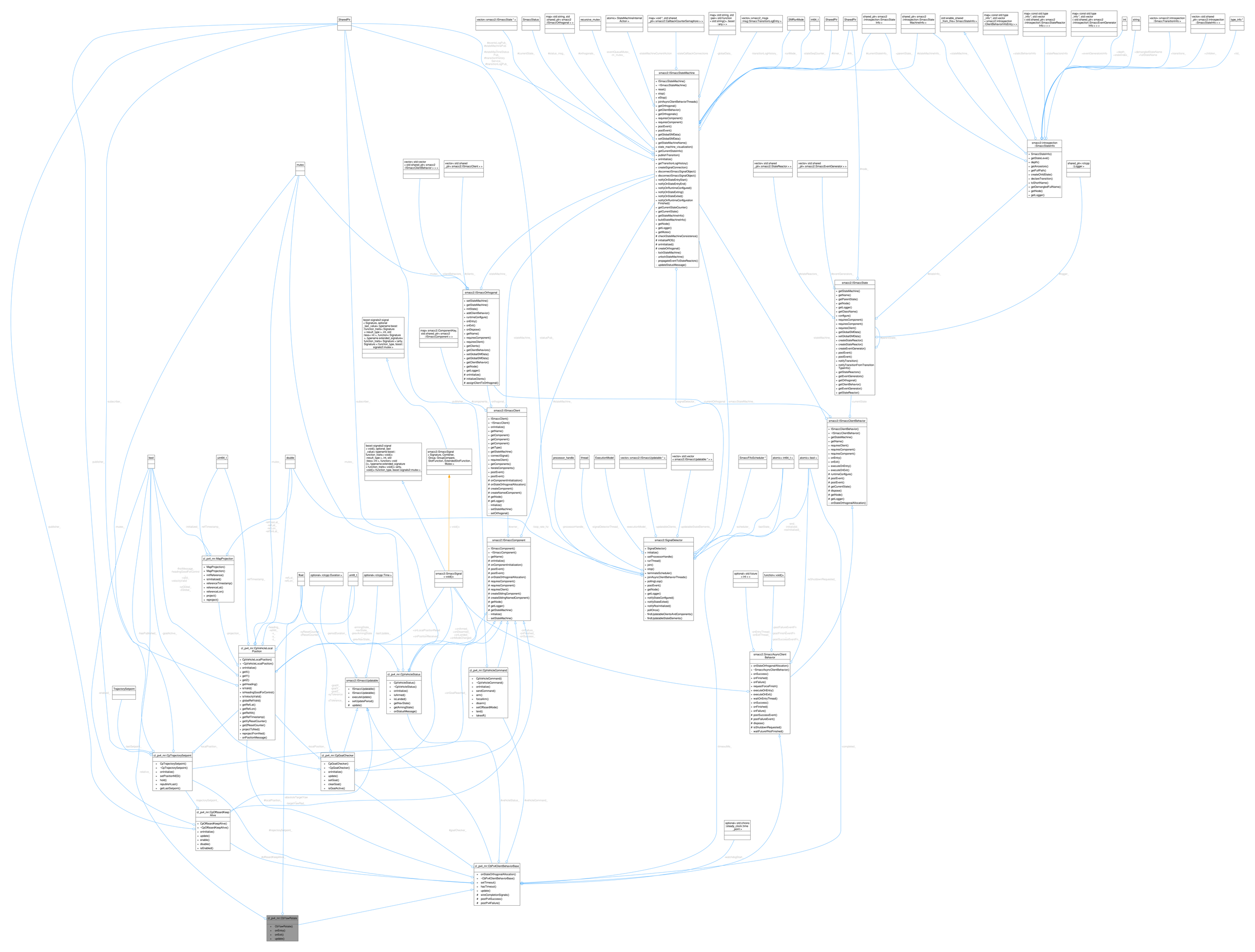
SharedPtr
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
SharedPtr
shared_ptr< smacc2::introspection::SmaccStateInfo >
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorVirtualEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
int
string
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
#eventLogPub_
#stateMachineStatusPub_
#transitionHistory_
#transitionLogPub_
#currentState_
#status_msg_
#orthogonals_
eventQueueMutex_
+n_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
-timer_
#ifsm_
#currentStateInfo_
+parentState_
#stateMachineInfo_
+stateMachine_
+staticBehaviorInfo
+stateReactorInfo
+eventGeneratorInfo
-depth_
+stateIndex_
+demangledStateName
+unStateName
+transitions_
+children_
+id_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::IStateReactor > >
vector< std::shared _ptr< smacc2::ISmaccEventGenerator > >
#node_
mutex
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
vector< std::shared _ptr< smacc2::ISmaccClient > >
-mutex_
-clientBehaviors_
#clients_
-stateMachine_
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# oninitialize()
# initializeClients()
# assignClientToOrthogonal()
boost::signals2::signal < Signature, optional _last_value= typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-publisher_
-subscriber_
-statusPub_
-signalDetector_
-currentOrthogonal_
-smaccStateMachine_
#stateMachine_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# orStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
-stateReactors_
-eventGenerators_
-stateInfo_
-logger_
-requiresInitialize_
-currentState
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
smacc2::Smacc Signal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
SmaccFifoScheduler *
atomic< int64_t >
atomic< bool >
smacc2::ISmaccClientBehavior
+ tISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponents()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
-executeMs_
-completed_
bool
uint64_t
double
-mutex_
-publisher_
-subscriber_
-initialized_
-refTimestamp_
-enabled_
-hasPublished_
-goalActive_
-projection_
-lastSetpoint_
-localPosition_
-relative_
-trajectorySetpoint_
-offboardKeepAlive_
-goalChecker_
-vehicleStatus_
-vehicleCommand_
-loop_rate_hz_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
-scheduler_
-lastState_
-end_
-initialized_
-isInitialized_
-isShutdownRequested_
-watchdogStart_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# orStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
optional< std::future < int > >
function< void()>
-onEntryThread_
-onExitThread_
-postFailureEventFn
-postFinishedEventFn
-postSuccessEventFn
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
float
optional< rclcpp::Duration >
uint8_t
optional< rclcpp::Time >
-refLat_
-refLon_
-heading
-refAlt
-x_
-y_
-z_
-xyResetCounter
-zResetCounter
-periodDuration_
-armingState_
-navState_
-prevArmingState
-prevNavState_
-lastUpdate_
+onLocalPositionReset
+onPositionReceived
+onArmed
+onDisarmed
+onFlightModeChanged
-onFailure
-onFinished
-onSuccess_
cl_px4_mr::CpVehicleLocal Position
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
+ onPositionMessage()
cl_px4_mr::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
-firstMessage_
-headingGoodForControl_
-valid_
-velocityValid_
-xyGlobal_
-zGlobal_
-refTimestamp_
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
-goalX_
-goalY_
-goalZ_
-xyTolerance_
-zTolerance_
cl_px4_mr::CpVehicleStatus
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
cl_px4_mr::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
TrajectorySetpoint
cl_px4_mr::CpTrajectorySetpoint
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
cl_px4_mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
-absoluteTargetYaw
-targetYawRad_
cl_px4_mr::CpOffboardKeep Alive
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
optional< std::chrono ::steady_clock::time _point >
cl_px4_mr::ClPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~ClPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
cl_px4_mr::ClbYawRotate
+ ClbYawRotate()
+ onEntry()
+ onExit()
+ update()